STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-REPRESENTATIONS-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 563-578 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 580-623, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_examples-defos.json (pb=1584, stmt 563-578, proof 580-623).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DVE: sources/stacks-examples-defos/pinned-original-source.tex, source lines 623–623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DVF: sources/stacks-examples-defos/pinned-original-source.tex, source lines 623–623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D3J: sources/stacks-examples-defos/pinned-original-source.tex, source lines 483–494. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DVQ: sources/stacks-examples-defos/pinned-original-source.tex, source lines 430–439. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0D3J let $\rho_0 : \Gamma \to GL_k(V)$ be a finite dimensional representation. Then*

$$T\mathcal{D}ef_{V,\rho_0} = \mathrm{Ext}^1_{k[\Gamma]}(V,V) = H^1(\Gamma, End_k(V)) \quad \textit{and} \quad Inf(\mathcal{D}ef_{V,\rho_0}) = H^0(\Gamma, End_k(V))$$

Thus $Inf(\mathcal{D}ef_{V,\rho_0})$ is always finite dimensional and $T\mathcal{D}ef_{V,\rho_0}$ is finite dimensional if Γ is finitely generated.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We first deal with the infinitesimal automorphisms. Let $M = V \otimes_k k[\epsilon]$ with induced action $\rho'_0 : \Gamma \to \mathrm{GL}_n(M)$. Then an infinitesimal automorphism, i.e., an element of $\mathrm{Inf}(\mathcal{D}ef_{V,\rho_0})$, is given by an automorphism $\gamma = \mathrm{id} + \epsilon\psi : M \to M$ as in the proof of Lemma 0DVQ, where moreover ψ has to commute with the action of Γ (given by ρ_0). Thus we see that

$$\mathrm{Inf}(\mathcal{D}ef_{V,\rho_0}) = H^0(\Gamma, \mathrm{End}_k(V))$$

as predicted in the lemma.

Next, let $(k[\epsilon], M, \rho)$ be an object of $\mathcal{F}$ over $k[\epsilon]$ and let $\alpha : M \to V$ be a Γ-equivariant map inducing an isomorphism $M/\epsilon M \to V$. Since M is free as a $k[\epsilon]$-module we obtain an extension of Γ-modules

$$0 \to V \to M \xrightarrow{\alpha} V \to 0$$

We omit the detailed construction of the map on the left. Conversely, if we have an extension of Γ-modules as above, then we can use this to make a $k[\epsilon]$-module structure on M and get an object of $\mathcal{F}(k[\epsilon])$ together with a map α as above. It follows that

$$T\mathcal{D}ef_{V,\rho_0} = \mathrm{Ext}^1_{k[\Gamma]}(V, V)$$

as predicted in the lemma. This is equal to $H^1(\Gamma, \mathrm{End}_k(V))$ by Étale Cohomology, Lemma 0DVE.

The statement on dimensions follows from Étale Cohomology, Lemma 0DVF. $\square$